Name: _____ ID: _____ Signature: _____

5. (15 points) You have simulated two systems (System No:1 and System No:2) for 5 replications each. The **sample means** for response times from System No:1 are 106, 95, 98, 85, 90, and 103 milliseconds, and from System No:2 they are 107, 108, 109, 108, 101, and 105 milliseconds. Can you, with 90% confidence (also 90% confidence), state that System No:2 has lower response time than System No:1? Why or why not? Show your work.

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4. (20 points) Consider the following single-server queuing system from time = 0 to time = ~~15~~ 20 sec. Arrivals and service times are

- Customer #1 arrives at t = 1 second and requires 5 seconds of service time
- Customer #2 arrives at t = 1 second and requires 2 seconds of service time
- Customer #3 arrives at t = 2 seconds and requires 3 seconds of service time
- Customer #4 arrives at t = 12 seconds and requires 6 seconds of service time

Determine the following measurements for the system for the given time period **(Carefully show your work including all figures and formulas. Credit will be given only to the complete solutions together with right formulas.)**

- Average number of the customers in the system,
- Average number of the customers in the queue,
- Average time spent in the system per customer,
- Average time spent in the queue per customer,
- Average arrival rate of the customers
- Average service rate of the system
- Utilization

Name: _____ ID: _____ Signature _____

**E381 Modeling and Discrete Simulation, Final Exam, 13 January 2014:**

(20 points) A simulation will be conducted for a car washing facility where various car models be included in the simulation. The facility manager says that service time is related with the e of the car. (It means that in the simulation, car size and service time might not be generated arately from different distributions).Therefore, data were collected for the next 10 cars.

| Car size (length in meters) | 4.65 | 4.36 | 4.78 | 4.88 | 4.80 | 4.42 | 4.88 | 4.9 | 4.35 | 4.07 |
|---|---|---|---|---|---|---|---|---|---|---|
| Washing Time (minutes) | 6.1 | 5.3 | 7.3 | 8.2 | 7.4 | 5.8 | 8.4 | 9.1 | 5.1 | 4.6 |

ute the sample correlation between the car size and the washing time and find out if the g time is dependent to car size?

Name: _____ ID: _____ Signature _____

**IE381/3081 Modeling and Discrete Simulation, Makeup Exam, 08 June 2017:**

**1. (30 points)** Random-variate generation

**a) (15 points)** Develop a random-variate generator for X with

$$cdf = \begin{cases} \frac{e^{2x}}{3}, & -\infty < x < 0 \\ 1 - e^{-3x}, & 0 < x < \infty \end{cases}$$

Generate 3 random variates for the given random values 0.034, 0.701, 0.4877

**b) (15 points)** Following random number sequence has been generated by a pseudo random number generator.

**0.1178, 0.5673, 0.7269, 0.4364, 0.8136, 0.3671, 0.6429, 0.9167, 0.4587, 0.7071, 0.3908, 0.6729**

Generate three Poisson variates with a ***mean*** $\alpha$***=1.5*** using the ***Acceptance-Rejection Technique***. Remember that Poisson random variable has probability mass function (pmf).

$$p(n) = P(N = n) = \frac{e^{-\alpha}\alpha^n}{n!}, n = 0, 1, 2, \ldots$$

mean = 1.5

**IE381 Modeling and Discrete Simulation, Makeup Exam, 06 January 2017**

**1. (40 points)** A modeler is building a model for the call center of Student Affairs Department. In the validation phase, output results of the simulation model will be compared with the observed system behavior which has ***a mean number of 73*** calls per day. 6 independent replications of the model were run and the following results are obtained as the average number of calls

| Replication $i$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Sample mean for replication $i$ | 73 | 66 | 63 | 60 | 64 | 70 |

**a) (25 points)** Develop and conduct a statistical test to evaluate whether model output is consistent with system behavior for the significance levels $\alpha$=0.10, $\alpha$=0.05 and $\alpha$=0.01

**b) (10 points)** What is the **power of this test** if the 5 calls is viewed as critical (i.e. at most 5 calls) for the significance level $\alpha$=0.05?

**c) (5 points)** What sample size is needed to have a power of 0.9 or more (i.e. at most 5 calls critical) for the significance level $\alpha$=0.05?

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at you have developed the model of a software development company with two system
em design there are two developers working in a team and in the second system design
ers working in a team. When the two systems are simulated for 8 replications, the
me to complete a work (in terms of hours) is given below. In each system design, same
ed for the same events

cond system is better than the first one (in terms of time to complete a work) for the
nd 99%?

| m 2 |
|---|
| 209 |
| 215 |
| 202 |
| 203 |
| 193 |
| 01 |
| 01 |
| 99 |

Name: _____ ID: _____ Signature _____

3. (25 points) Assume that you have developed the model of a software development co
designs. In the first system design there are two developers working in a team and in the
there are three developers working in a team. When the two systems are simulate
sample means for the time to complete a work (in terms of hours) is given below. In ea
random numbers are used for the same events.

Find out whether the second system is better than the first one (in terms of time to
confidence levels 98% and 99%?

| Rep.# | System 1 | System 2 |
|---|---|---|
| 1 | 205 | 209 |
| 2 | 210 | 215 |
| 3 | 208 | 202 |
| 4 | 199 | 203 |
| 5 | 207 | 193 |
| 6 | 211 | 201 |
| 7 | 206 | 201 |
| 8 | 205 | 199 |

**2. (35 points)** Assume that the model defined in ***Problem 1*** has been validated. Now, the modeler is testing system performance for new system designs. For a new design, the simulation analyst made **R=10** replications. Sample mean of number of calls (successfully replied) for each replication is given in the table below.

| Replication $r$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Sample mean for replication $r$ | 88 | 86 | 80 | 75 | 65 | 75 | 69 | 81 | 77 | 79 |

**a) (20 points)** Estimate the long run average number of calls and find the error of estimate by 95% and 99% confidence intervals (find the confidence intervals).

Name: _____ ID: _____ Signature _____

**2.** (25 points)

**a)** (15 points) Given an M/M/1 queue with an arrival rate of 5 jobs per second and a service rate of 10 jobs per second what is the mean number of customers in the system? What is the mean customer delay?

$\rho = \lambda/\mu$ , $P_n = (1-\rho)\rho^n$

$L = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1-\rho}$

**b)** (10 points) Repeat (b) for an M/G/1 queue?

$L = \rho + \frac{\rho^2 (1 + \sigma^2 \mu^2)}{2(1-\rho)}$

**3.** (20 points) A two-runway (one runway for landing, one runway for taking off) airport is being designed for planes. The time to land an airplane is known to be exponentially distributed, with a mean of 2 minutes. If airplane arrivals are assumed to occur at random, what arrival rate can be tolerated if the average wait in the sky is not to exceed 5 minutes?